Quantum coherence in the localization transition

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A dynamical signature of localization in quantum systems is the absence of transport which, in turn, is governed by the amount of coherence that configuration space states possess with respect to the Hamiltonian eigenbasis. To make this observation precise, we study the localization transition via quantum coherence measures, averaged over the Hamiltonian eigenstates. We show that the (average) escape probability, which is known to show distinct behavior in the ergodic and localized phases, can be cast as a coherence measure and, conversely, coherence measures can probe localization. For infinitesimal perturbations of the Hamiltonian, one can resort to a differential approach to coherence, for which we determine the associated Riemannian metric and show that it can be mapped onto dynamical conductivity, a quantity of experimental relevance which is known to have a distinctively different behavior between the ergodic and the many-body localized phases. We provide supporting numerical evidence for Anderson and many-body localization.

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I. INTRODUCTION

One of the conceptual pillars of quantum theory is the superposition principle and, directly arising from it, the notion of *quantum coherence* [1]. A quantum state is deemed to be coherent with respect to a complete set of states if it can be expressed as a non-trivial linear superposition of these states. Recently, there has been an effort to formulate a *quantitative* theory of quantum coherence [2–4]. The focus of this theory has been quantum information processing tasks, since generating and preserving quantum coherence constitutes one of the essential prerequisites.

In this work, we utilize measures that arose from this information-theoretic perspective on coherence to study phase transitions in quantum many-body systems. More specifically, we focus on systems in which a radical change is occurring at the level of whole Hamiltonian eigenstate system as opposed, e.g., to the ground-state only. A prominent example of these "infinite temperature" or "eigenstate" phase transitions is provided by the phenomena of Anderson [5, 6] and Many-Body Localization (MBL) [7–9]. The latter transitions are going to be the focus of the present work.

A conceptual connection between quantum coherence and the transition of a quantum system from an ergodic phase to a localized one is straightforward. One of the signatures of localization is the absence of transport, with respect to some properly defined positional degree of freedom. On the other hand, transport properties are governed by the coherence between the Hamiltonian eigenbasis and the positional one. Hence one should expect an abrupt change in the coherence properties of the Hamiltonian eigenvectors at the transition point.

Here we make the above intuition quantitative by invoking measures of quantum coherence, defined with respect to the energy eigenbasis, averaged over a complete set of position states. Such quantities essentially capture the difference between two complete orthonormal sets of eigenstates associated with two hermitian operators [10, 11]. We first show that a well-studied quantity in localization, the *escape probability* (or, equivalently, the *second participation ratio*) can be expressed directly as a coherence average. We then argue that broad families of coherence measures, arising from the the resource-theoretic perspective, can be used to define an "order parameter" for localization. We also provide supporting numerical evidence to this claim. Moreover, we establish a straightforward connection between the differential-geometric version of our average coherence and dynamical conductivity, an experimentally accessible quantity, which is known to behave differently in the ergodic and MBL phases [12].

This paper is organized as follows. In section II we introduce measures of coherence for quantum states, explain how one can average coherence over a complete set of states in order to obtain a quantifier of coherence for quantum operations, and connect with the associated escape probability. In section III we study the behavior of two of the introduced measures in the Anderson localization transition, while in section IV we do the same for a many-body system that exhibits a transition to a MBL phase. In section V we turn to the Riemannian metric that results from the average coherence between bases that differ infinitesimally and relate with the MBL transition. Finally, in section VI we conclude. Proofs can be found in section A of the Appendix.

II. QUANTUM COHERENCE OF STATES AND OPERATIONS

A. Coherence of states

Consider a quantum system, described by a finite dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^d$. A state $|\psi\rangle \in \mathcal{H}$ is deemed *coherent* with respect to a fiducial orthonormal

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basis $\{|i\rangle\}_{i=1}^d$ if the expansion $|\psi\rangle = \sum_i a_i |i\rangle$ contains more than one non-vanishing term, otherwise it is called *incoherent*. This notion extends straightforwardly to the set of density operators $\mathcal{S}(\mathcal{H})$. Any $\rho \in \mathcal{S}(\mathcal{H})$ is regarded as *coherent* with respect to the preferred basis if the corresponding matrix ρ_{ij} has non-trivial off-diagonal elements, otherwise it is *incoherent*.

Quantum coherence is usually defined relative to a reference basis. In fact, one needs a weaker notion than that of a basis, since phase degrees of freedom and ordering of an orthonormal basis $\{|i\rangle\}_{i=1}^d$ are physically redundant. In other terms, bases differing by transformations of the form $|j\rangle \mapsto e^{i\theta_j} |\pi(j)\rangle$ ($\pi \in S_d$ is a permutation) are equivalent as far as coherence is concerned. The relevant object, taking into account this freedom, is a complete set of orthogonal, rank-1 projection operators $B = \{P_i\}_{i=1}^d$, where $P_i := |i\rangle\langle i|$. In the rest of this work, we will refer for convenience to the set B itself as a "basis".

While all states non-diagonal in B carry coherence, some of them might resemble incoherent states more than others. This notion is made precise by the introduction of (B-dependent) functionals $c_B : \mathcal{S}(\mathcal{H}) \to \mathbb{R}_0^+$ that are said to quantify coherence [3]. Quantifiers of coherence satisfy $c_B(\rho_{\text{inc}}) = 0$ for all B-incoherent states (i.e., states diagonal in B). In this work, we will focus for concreteness on the *2-coherence* and the *relative entropy of coherence*, defined respectively by

$$c_B^{(2)}(\rho) := \|(\mathcal{I} - \mathcal{D}_B)\,\rho\|_2^2 = \sum_{i\neq j} |\rho_{ij}|^2 \tag{1a}$$

$$c_B^{(\text{rel})}(\rho) := S\left[\mathcal{D}_B\left(\rho\right)\right] - S\left(\rho\right) \ , \tag{1b}$$

where we have introduced the B-dephasing superoperator

$$\mathcal{D}_B(X) := \sum_{i=1}^d P_i X P_i \ , \tag{2}$$

S above denotes the usual von-Neumann entropy $S(\rho) := -\operatorname{Tr}\left(\rho \log(\rho)\right)$ and the (Schatten) 2-norm of an operator X is defined as $\|X\|_2 := \sqrt{\operatorname{Tr}\left(X^\dagger X\right)}$. Relative entropy of coherence is a central measure in the resource theories of coherence and admits an operational interpretation, e.g., as a conversion rate of information-theoretic protocols [13, 14]. The 2-coherence admits an interpretation as an escape probability, as will be shown momentarily.

B. Coherence of unitary quantum processes via probabilistic averages

In this section we discuss how, given a coherence measure c_B, one can define a corresponding notion of coherence for a unitary superoperator $\mathcal{U}$ via probabilistic averages. We refer to this approach to quantification of coherence for quantum operations as *Coherence-Generating Power* (CGP) [10, 15–17].

Consider a basis $B = \{P_i\}_{i=1}^d$ and define a probabilistic ensemble of incoherent states, i.e., a random variable $\rho_{\text{inc}}(\boldsymbol{p}) = \sum_i p_i P_i$, where $\{p_i\}_i$ ($p_i \geq 0$, $\sum_i p_i = 1$) are random and distributed according to a prescribed measure $\mu(\boldsymbol{p})$. Then, one can define the CGP

$$C\left(\mathcal{U}, c_B, \mu\right) := \int d\mu(\boldsymbol{p})\, c_B\left[\mathcal{U}\left(\rho_{\text{inc}}(\boldsymbol{p})\right)\right] \tag{3}$$

which characterizes the average effectiveness of the quantum process $\mathcal{U}$ to generate coherence out of random incoherent states in B. Since the unitary $\mathcal{U}(X) = UXU^\dagger$ can be thought of as connecting the bases B and $B' = \{\mathcal{U}\left(P_i\right)\}_i$, one can also interpret $C\left(\mathcal{U}, c_B, \mu\right)$ as the average coherence with respect to B of a random state which is incoherent in B'.

Without any additional structure, it is a natural choice to consider averaging only over pure states with equal weight over each of them , i.e., take

$$\mu_{\text{unif}}(\boldsymbol{p}) := \frac{1}{d} \sum_i \delta\left(\boldsymbol{p} - \boldsymbol{e}_i\right) \tag{4}$$

where $(\boldsymbol{e}_i)_j := \delta_{ij}$ [18]. This choice directly leads to the expression

$$C\left(\mathcal{U}, c_B, \mu_{\text{unif}}\right) = \frac{1}{d} \sum_{i=1}^d c_B\left[\mathcal{U}\left(P_i\right)\right] \ . \tag{5}$$

We now simplify Eq. (5) when the coherence measure is the 2-coherence or the relative entropy of coherence, namely for

$$C_B^{(2)}\left(\mathcal{U}\right) := C\left(\mathcal{U}, c_B^{(2)}, \mu_{\text{unif}}\right) \ , \tag{6a}$$

$$C_B^{(\text{rel})}\left(\mathcal{U}\right) := C\left(\mathcal{U}, c_B^{(\text{rel})}, \mu_{\text{unif}}\right) \ . \tag{6b}$$

Proposition 1. Let $B = \{P_i\}_{i=1}^d$ be a basis, $\mathcal{U}$ a unitary quantum process and $X_{\mathcal{U}}$ denote the bistochastic matrix with elements $\left(X_{\mathcal{U}}\right)_{ij} := \operatorname{Tr}\left(P_i \mathcal{U}(P_j)\right)$. Then,

$$C_B^{(2)}\left(\mathcal{U}\right) = 1 - \frac{1}{d} \operatorname{Tr}\left(X_{\mathcal{U}}^T X_{\mathcal{U}}\right) \ . \tag{7}$$

and

$$C_B^{(\text{rel})}\left(\mathcal{U}\right) = H(X_{\mathcal{U}}) \ , \tag{8}$$

where $H(X) := -\frac{1}{d} \sum_{i,j} X_{ij} \log(X_{ij})$ denotes the generalization of the Shannon entropy over bistochastic matrices.

The two CGP quantities are related as

$$C_B^{(\text{rel})} \geq -\log\left(1 - C_B^{(2)}\right) \ . \tag{9}$$

The inequality follows from the above Proposition, together with the concavity of the logarithmic function.

C. Coherence and escape probability

Let us consider a finite-dimensional quantum system whose dynamics is specified by a Hamiltonian H. Suppose the system is initialized in a state $|\psi\rangle$ and one is interested in the *escape probability*

$$\mathcal{P}_\psi := 1 - \overline{|\langle\psi|e^{-iHt}|\psi\rangle|^2} \ , \tag{10}$$

where the overline denotes the infinite time-average

$$\overline{f(t)} := \lim_{T\to\infty} \frac{1}{T}\int_0^T dt\, f(t) \ . \tag{11}$$

For instance, in the case of a particle hopping on a lattice which is initialized over a single site, $\mathcal{P}_\psi$ corresponds to the average probability of the particle escaping the initial site.

At this point, let us note that in finite dimensions observable quantities such as $\langle A(t)\rangle := \mathrm{Tr}[A(t)\rho_0] = \mathrm{Tr}[A\rho(t)]$ do not converge to any limit as $t\to\infty$. Instead, for large enough systems sizes, such quantities start from an initial value and then start oscillating around a value given by $\overline{\langle A(t)\rangle}$ [19–22]. Note that if a function $f(t)$ has a limit for $t\to\infty$, this limit must coincide with $\overline{f(t)}$. Hence, the infinite time average provides a way to extract the infinite time limit even when the latter strictly speaking does not exist.

If the Hamiltonian in consideration has non-degenerate energy gaps, then the quantity $(1-\mathcal{P}_\psi)^{-1}$ provides a sufficient condition for the state to equilibrate [19, 20], in the sense of observables having small temporal fluctuations around the mean value. Since many-body localization is a mechanism by which quantum systems can escape equilibration, it is perhaps no surprise that the effective dimension is related to the localization transition (see the Appendix B for more details on related quantities).

The following Proposition establishes the fact the the 2-coherence of a state, quantified with respect to the Hamiltonian eigenbasis, is the time-averaged escape probability of the state.

Proposition 2. Let $H = \sum_i E_i |\phi_i\rangle\langle\phi_i|$ be a non-degenerate Hamiltonian.

(i) For any state $|\psi\rangle$,

$$\mathcal{P}_\psi = c_B^{(2)}(|\psi\rangle\langle\psi|) \ , \tag{12}$$

where $B = \{|\phi_i\rangle\langle\phi_i|\}_i$ is the eigenbasis of the Hamiltonian.

(ii) Denote the escape probability averaged over a set of orthonormal states $B' = \{|i\rangle\langle i|\}_{i=1}^d$ as

$$\mathcal{P}_{B'} := \frac{1}{d}\sum_{i=1}^d \mathcal{P}_i \ . \tag{13}$$

Then,

$$\mathcal{P}_{B'} = C_B^{(2)}(\mathcal{V}) = C_{B'}^{(2)}(\mathcal{V}^\dagger) \ , \tag{14}$$

where $B = \{|\phi_i\rangle\langle\phi_i|\}_i$ is the eigenbasis of the Hamiltonian and $V = \sum_i |i\rangle\langle\phi_i|$ is the intertwiner between B and B'.

The last equation above demonstrates that the role of the bases B and B' can be interchanged. For instance, one can equivalently think in terms of the average coherence over Hamiltonian eigenstates, quantified with respect to the position basis.

The above identification also gives a physical interpretation to the 2-coherence and the associated CGP. More importantly, the escape (or return) probability is a well-known measure in the theory of localization [23, 24] and the fact that it can be thought of as coherence gives rise to the question: can other measures arising from the resource theoretic framework of coherence give rise to probes of localization in a similar manner? To demonstrate that this is indeed the case, we consider the cases and Anderson and MBL, where we study the two introduced CGP quantities.

III. COHERENCE-GENERATING POWER AND LOCALIZATION IN THE 1-D ANDERSON MODEL

The Anderson model [23] in one dimension is described by the Hamiltonian

$$H_W = -\sum_{i=1}^L (|i\rangle\langle i+1| + |i+1\rangle\langle i|) + \sum_{i=1}^L \epsilon_i |i\rangle\langle i| \tag{15}$$

over L sites (i.e., $d = L$) with periodic boundary conditions, where the on-site energies ϵ_i are independent and identically distributed (i.i.d.) random variables and follow a uniform distribution of width $2W$. It is known that the model is localized for any degree of disorder $W > 0$ [25].

Since localization can be dynamically characterized by the absence of transport, a notion referring to the "position" basis $B' = \{|i\rangle\langle i|\}_{i=1}^L$ in Eq. (15), we consider coherence quantified with respect to that basis. Let us examine the behavior of functionals $C_B(\mathcal{V}_W)$ where the unitary V_W is the intertwiner between B' and the eigenbasis of H_W. In fact, Proposition 2 immediately implies that $\langle C_B^{(2)}(\mathcal{V}_W)\rangle$ is a probe to localization ($\langle\cdot\rangle$ denotes averaging over disorder). More specifically, localization implies that in thermodynamic limit the (averaged over disorder) return probability in the localized phase is non-vanishing, i.e., $\lim_{L\to\infty} \langle\overline{|\langle j|e^{-iH_W t}|j\rangle|^2}\rangle > 0$ for any $W > 0$. In turn, this is equivalent to $\mathcal{P}_j < 1$ (in the thermodynamic limit) for all sites j, hence also

$$\lim_{L\to\infty} \langle C_B^{(2)}(\mathcal{V}_W)\rangle < 1 \tag{16}$$

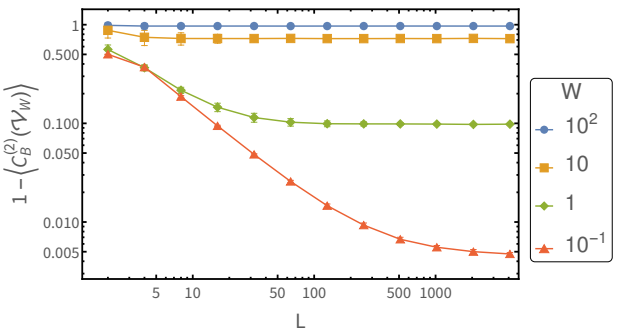

FIG. 1. Log-log plot of the average return probability $1 - \langle C_B^{(2)}(\mathcal{V}_W)\rangle$ as a function of the system size L for different values of the disorder strength W. The system is in the localized phase for all $W > 0$, since the asymptotic escape probability is strictly less that 1 for $L \to \infty$. The number of realizations range from 30 000 for small sizes to just 8 for the largest size. Error bars represent standard deviations.

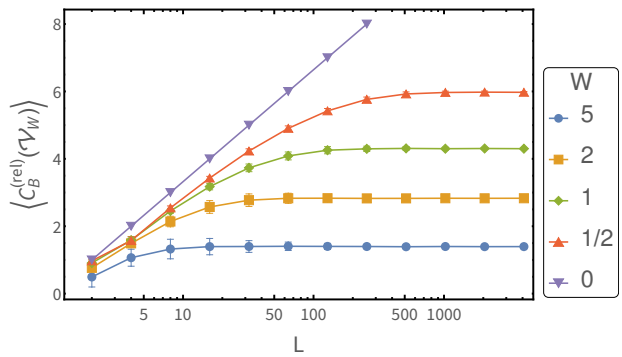

FIG. 2. Log-linear plot of $\langle C_B^{(\mathrm{rel})}(\mathcal{V}_W)\rangle$ as a function of the system size L for different values of the disorder strength W. The system is in the localized phase for all $W > 0$, in which the asymptotic value is finite. In the ergodic phase ($W = 0$) $\langle C_B^{(\mathrm{rel})}(\mathcal{V}_W)\rangle$ diverges logarithmically. The number of realizations range from 30 000 for small sizes to just 8 for the largest size. Error bars represent standard deviations. Entropy has logarithm with base 2.

by Eq. (13). Notice that H_W for $W > 0$ is generically non-degenerate so Proposition 2 applies. We verify this claim by numerical simulations (see Figure 1).

The Hamiltonian $H_{W=0}$ is degenerate in the ergodic phase, hence the intertwiner $\mathcal{V}_{H_{W=0}}$ is not well-defined. Nevertheless, as we show in Appendix C, for any choice of eigenbasis of H_W it holds that

$$\lim_{L\to\infty} C_B^{(2)}(\mathcal{V}_{W=0}) = 1 \ , \tag{17}$$

hence the coherence average $\langle C_B^{(2)}(\mathcal{V}_W)\rangle$ unambiguously distinguishes the two behaviors.

The role of the quantity $C_B^{(2)}(\mathcal{V}_W)$ might seem special as a probe to localization due to its interpretation as average escape probability. In fact, other measures, arising from an information-theoretic viewpoint to coherence, have analogous properties. Let's now consider the relative entropy CGP of the intertwiner, namely $C_B^{(\mathrm{rel})}(\mathcal{V}_W)$. It's asymptotic value $L \to \infty$ as a function of the disorder strength W is plotted in Figure 2. In the ergodic phase $W = 0$ it diverges logarithmically

$$C_B^{(\mathrm{rel})}(\mathcal{V}_{W=0}) \sim \log(L) \ . \tag{18}$$

This can be easily verified analytically for an intertwiner connecting two mutually unbiased bases, i.e., for $|(V_{W=0})_{ij}| = |\langle i|\phi_j\rangle| = 1/L$ for all i, j. In that case Eq. (18) holds with equality, as it directly follows from Proposition 1. In Appendix C we show that the result again holds in the thermodynamic limit independently of the specific choice for the intertwiner.

We now provide a non-rigorous argument to relate the averages $\langle C_B^{(2)}(\mathcal{V}_{W>0})\rangle$ and $\langle C_B^{(\mathrm{rel})}(\mathcal{V}_{W>0})\rangle$ to the corresponding localization lengths. In the localized phase, the eigenvectors typically decay exponentially, i.e.,

$$|\langle i|\phi_j\rangle|^2 \leq c_j \exp\left(-|i-\alpha_j|/\xi_j\right) \ , \tag{19}$$

where α_j is the site around which $|\phi_j\rangle$ is localized, while due to the periodic boundary conditions $|i-\alpha_j|$ above should be understood as $\min(|i-\alpha_j|, |i-\alpha_j \pm L|))$. If one uses the ansatz

$$(X_{\mathcal{V}_W})_{ij} = c_j \exp\left(-|i-\alpha_j|/\xi_j\right) \ , \tag{20}$$

then for $L \gg 1$

$$\langle C_B^{(2)}(\mathcal{V}_{W>0})\rangle \cong 1 - \frac{1}{L}\sum_j \frac{\tanh^2[(2\xi_j)^{-1}]}{\tanh(\xi_j^{-1})} \tag{21a}$$

and

$$\langle C_B^{(\mathrm{rel})}(\mathcal{V}_{W>0})\rangle \cong \frac{1}{L}\sum_{j=1}^{L}\left([\xi_j \sinh(1/\xi_j)]^{-1} - \ln\left(\tanh\left[(2\xi_j)^{-1}\right]\right)\right) \tag{21b}$$

(entropy here has natural logarithm). A detailed derivation can be found in Appendix D.

The expression (21b) for $\xi_j \gg 1$ can be expanded as $\langle C_B^{(2)}(\mathcal{V}_\Gamma)\rangle = \frac{1}{L}\sum_{j=1}^{L}\left(1 + \ln(2\xi_j) + O(\xi_j^{-2})\right)$, which is consistent with the numerically observed behavior that it remains finite in the localized phase while it diverges logarithmically in the ergodic one.

The accuracy of equations (21) can be assessed by comparing with cases for which an analytical expression can be obtained for the localization lengths ξ_j as a function of the disorder strength. We now consider such a case, described by a Hamiltonian as in Eq. (15), but with on-site energies that follow a Cauchy distribution with parameter Γ and vanishing mean (also known as Lloyd model [26]). We focus for concreteness on Eq. (21a) and we denote the corresponding Hamiltonian and intertwiner as H_Γ and $\mathcal{V}_\Gamma$, respectively. Utilizing a well-known result

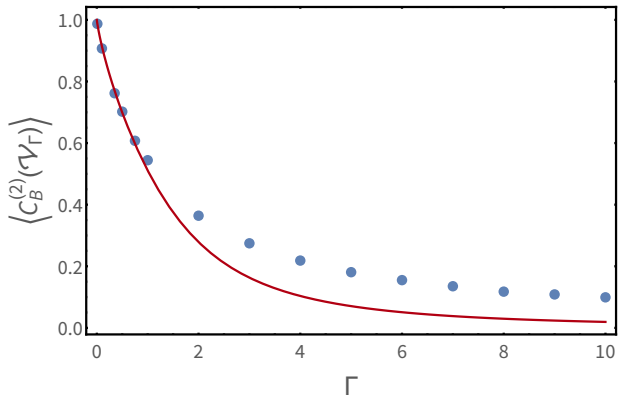

FIG. 3. Plot of the escape probability $\langle C_B^{(2)}(\mathcal{V}_\Gamma)\rangle$ as a function of the disorder strength Γ for the Lloyd model Hamiltonian H_Γ, as predicted analytically by the heuristic Eq. (21a) (solid line) and the numerical simulations (points). For the case of the numerical simulation, $L \to \infty$ is extrapolated by averaging over disorder for sizes up to $L = 2^{12}$. Standard deviations are within the point radius.

from Thouless [27] that connects the localization length with the energy spectrum, one can express the RHS of Eq. (21a) as a function of the disorder strength Γ. This allows for a direct comparison with numerical evaluations of the mean $\langle C_B^{(2)}(\mathcal{V}_\Gamma)\rangle$, depicted in Figure 3, yielding a sound agreement for small disorder ($\Gamma < 1$). We present the details in section E of the Appendix.

IV. COHERENCE-GENERATING POWER AND MANY-BODY LOCALIZATION

We now turn to a disordered quantum many-body system admitting a phase diagram with an ergodic phase at low enough disorder and an MBL phase at strong disorder. For this purpose, we consider a transverse-field Heisenberg spin-1/2 chain in a random magnetic field (along the $\hat{z}$-axis) over L sites ($d = 2^L$) with periodic boundary conditions, described by the Hamiltonian

$$H_{\text{XXX}} = \frac{1}{2}\sum_{i=1}^{L}\left[\sigma_i^x\sigma_{i+1}^x + \sigma_i^y\sigma_{i+1}^y + \sigma_i^z\sigma_{i+1}^z\right] + h_x\sum_{i=1}^{L}\sigma_i^x + \sum_{i=1}^{L} w_i\sigma_i^z \ , \tag{22}$$

where the h_x is the strength of the transverse field and the local field strengths are i.i.d. random variables with uniform distribution $w_i \in [-W, W]$. The model has been extensively studied numerically and is known to exhibit a transition from an ergodic to an MBL phase at disorder strength $W_C \approx 3.7$ (in the absence of the transverse field term), see [24, 28] and references therein.

As for the Anderson Hamiltonian, we study the behavior of the CGP $\langle C_B^{(2)}(\mathcal{V}_W)\rangle$ and $\langle C_B^{(\text{rel})}(\mathcal{V}_W)\rangle$, where $\mathcal{V}_W$ is the intertwiner between the Hamiltonian and the configuration space basis, which here is taken to be the product $\bigotimes_i \sigma_i^z$ eigenbasis. We find a distinct behavior of the quantities $\langle C_B^{(2)}(\mathcal{V}_W)\rangle$ and $\langle C_B^{(\text{rel})}(\mathcal{V}_W)\rangle$ between the ergodic and MBL phases of the model, as also expected from the numerical results in [29–31].

More specifically, for sizes up to $L = 14$, none of the studied CGP quantities seems to retain a constant asymptotic value as in the Anderson case. Nonetheless, the (average) *return probability*

$$P_{\text{return}} := 1 - \mathcal{P}_{B'} = 1 - \langle C_B^{(2)}(\mathcal{V}_W)\rangle \tag{23}$$

as a function of the number of spins L is consistent with an exponential decay model

$$P_{\text{return}} \propto 2^{-\lambda_W^{(2)} L} = d^{-\lambda_W^{(2)}} \ . \tag{24}$$

The extrapolated rates $\lambda_W^{(2)}$, plotted in Figure 4, are close to 1 in the ergodic phase, while they drop at the transition point, obtaining a significantly reduced value at the MBL phase. On the other hand, the relative entropy CGP is consistent with a model

$$\langle C_B^{(\text{rel})}(\mathcal{V}_W)\rangle = \lambda_W^{(\text{rel})} L + \text{const} \tag{25}$$

(see Figure 4), with a rate $\lambda_W^{(\text{rel})}$ that is close to 1 for small disorder and drops significantly in the MBL phase.

Finally, we comment on our findings from the typicality point of view. In [15] it was shown that, if the intertwiner is chosen at random from the unitary group $V \in U(d)$ according to the Haar measure, then $C_B^{(2)}$ is concentrated near its mean

$$\langle C_B^{(2)}(\mathcal{V})\rangle_V = 1 - \frac{2}{d+1} \tag{26}$$

($\langle\cdot\rangle_V$ denotes the Haar average over the intertwiner), with overwhelming probability for large Hilbert space dimension d (here B can be any fixed basis). In other words, the typical rate for P_{return} is $\lambda_{\text{Haar}}^{(2)} \approx 1$. From that perspective, an ergodic behavior is the typical one, while the MBL case can be seen as a highly atypical outlier.

V. DIFFERENTIAL GEOMETRY OF COHERENCE-GENERATING POWER AND MBL

In this section we study the behavior of the CGP $C_B^{(2)}(\delta\mathcal{V})$ when the intertwiner $\delta\mathcal{V}$ connects two bases that are "infinitesimally close" to each other. This results in a differential-geometric construction whose central quantity is a Riemannian metric. As we will show, the resulting metric (i) is directly connected to the dynamical conductivity, which is a quantity of experimental relevance, and (ii) behaves distinctly in the MBL and ergodic phases. The detailed mathematical structure is presented in Appendix F.

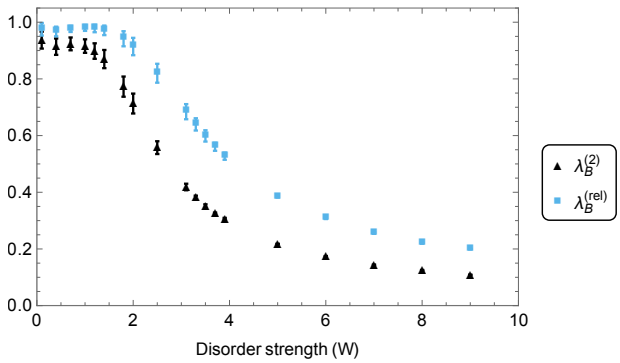

FIG. 4. Asymptotic behavior for the slope of $\log_2\left(1-\langle C_B^{(2)}(\mathcal{V}_W)\rangle\right)=\log_2\left(P_{\mathrm{return}}\right)$ and $\langle C_B^{(\mathrm{rel})}(\mathcal{V}_W)\rangle$ for large L as a function of the disorder strength W for H_{XXX} at $h_x=0.3$. The slope was extracted for sizes $L=4,6,\cdots,14$, with sample sizes $20\,000, 20\,000, 20\,000, 8000, 2000, 200$; except at the $W=3.7$, where the sample sizes were doubled. Entropy has logarithm with base 2.

Consider a complete orthonormal family of states $\{|\phi_i(\lambda)\rangle\}_{i=1}^d$, parametrized by a set of parameters λ. This is the relevant case, for instance, when one studies the eigenvectors associated with a family of Hamiltonians $H(\lambda)$. The infinitesimal adiabatic intertwiner $\delta\mathcal{V}$ is a unitary map defined by

$$\delta\mathcal{V}\left(|\phi_i(\lambda)\rangle\langle\phi_i(\lambda)|\right)=|\phi_i(\lambda+d\lambda)\rangle\langle\phi_i(\lambda+d\lambda)| \quad , \tag{27}$$

where $H(\lambda)\,|\phi_i(\lambda)\rangle=E_i(\lambda)\,|\phi_i(\lambda)\rangle$.

The CGP of $\delta\mathcal{V}$ has the form $C_B^{(2)}(\delta\mathcal{V})=2gd\lambda^2$, where g is a metric given by

$$g\coloneqq\frac{1}{d}\sum_{i=1}^{d}\chi_i \quad , \tag{28a}$$

$$\chi_i\coloneqq\left\langle\frac{\partial\phi_i}{\partial\lambda}\middle|\frac{\partial\phi_i}{\partial\lambda}\right\rangle-\left\langle\phi_i\middle|\frac{\partial\phi_i}{\partial\lambda}\right\rangle\left\langle\frac{\partial\phi_i}{\phi_i}\middle|\phi_i\right\rangle \quad , \tag{28b}$$

i.e., it is itself a mean of the metrics χ_i which are associated to the vectors $|\phi_i\rangle$. When the latter are Hamiltonian eigenstates, χ_i are known as *fidelity susceptibilities* [32–34] and the ground state susceptibility χ_0 plays a key role in the differential geometric approach to quantum phase transitions [35].

In order to connect with quantities of experimental relevance, let us now consider the thermal analog of the metric g. We denote $g_T=\sum_i p_i\chi_i$, where $p_i=\exp\left(-E_i/T\right)/Z$ are the thermal weights and Z denotes the partition function. The quantity g_T, defined in [36] as a generalization of the fidelity susceptibility at finite temperature ($g=g_{T=\infty}$), can be thought of as the metric associated with the thermal analogue of the CGP $C(\mathcal{V},c_B^{(2)},\mu_T)$, where the measure μ_T weights the Hamiltonian eigenstates with the associated Gibbs weights. The quantity g_T can be expressed via the (imaginary part of the) dynamical susceptibility $\chi_{VV}(\omega)$, where $V=\partial_\lambda H(\lambda)$. More precisely (see [36]),

$$g_T=\int_0^\infty\frac{d\omega}{\pi}\frac{\chi''_{VV}(\omega)}{\omega^2}\coth\left(\frac{\omega}{2T}\right). \tag{29}$$

The above formula is remarkable, as it demonstrates that the, apparently abstract, quantity $C_B^{(2)}(\delta\mathcal{V})$ is simply connected with a quantity measurable in experimental setups [37–39]. We also note that, although Eq. (29) is not straightforwardly applicable in the infinite temperature limit, in this limit one obtains

$$g=g_{T=\infty}=\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\sigma_{VV}(\omega)}{\omega^2}d\omega \quad , \tag{30}$$

where $\sigma_{VV}(\omega)$ is the high-temperature dynamical conductivity [40]. In this case, the role of g is played by the d.c. dielectric polarizability [12, 41].

The quantities g_T and g not only allow to make contact with experiments, but also have been studied in the context of thermalization and of MBL. In particular, it is believed that $g\to\infty$ in the thermodynamic limit, both for the ergodic and the sub-diffusive phase. Instead, in the MBL phase $g\to$ constant $<\infty$ [12]. In the light of Eq. (28), these results mean that the CGP of the adiabatic intertwiner between nearby Hamiltonians has distinctively different behaviors in the ergodic and in the MBL phases.

VI. CONCLUSIONS

In this work we have brought together ideas from quantum information and geometry, on the one hand, and the physics of disordered systems, on the other. We established a connection between the quantitative approach to coherence, originating from the perspective of resource theories [3, 4], and localization [7–9, 23].

More specifically, we studied the behavior over the ergodic, Anderson and many-body localized phases in terms of the scaling properties of coherence averages that are associated to the intertwiner connecting the Hamiltonian eigenvectors with the configuration space basis. Furthermore, we built an associated differential-geometric version for infinitesimal perturbations of the Hamiltonian, and showed that the resulting Riemannian metric can be mapped onto known physical quantities which have a sharply distinct behavior in the ergodic and in the MBL phases.

In this work, the focus was on two specific measures of coherence, but the framework laid is general and allows for an analogous construction to be carried over for any coherence measure. In fact, the two CGP quantities studied are proper monotones in the framework of incompatibility [11], where the latter is cast (for the case of orthogonal measurements) as a resource theory over the set of bases. It remains an open question to characterize the essential properties of coherence or incompatibility

quantifiers that may allow for an insightful connection with localization.

Establishing further, more rigorous, theoretical as well as numerical grounds for this connection between the information-theoretic approach to quantum coherence and many-body theory provides a direction for future research.

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Appendix A: Proofs of Propositions

Proposition 1. Let $B = \{P_i\}_{i=1}^d$ be a basis, $\mathcal{U}$ a unitary quantum process and $X_{\mathcal{U}}$ denote the bistochastic matrix with elements $(X_{\mathcal{U}})_{ij} := \mathrm{Tr}\left(P_i \mathcal{U}(P_j)\right)$. Then,

$$C_B^{(2)}(\mathcal{U}) = 1 - \frac{1}{d}\,\mathrm{Tr}\left(X_{\mathcal{U}}^T X_{\mathcal{U}}\right) \ . \tag{7}$$

and

$$C_B^{(\mathrm{rel})}(\mathcal{U}) = H(X_{\mathcal{U}}) \ , \tag{8}$$

where $H(X) := -\frac{1}{d}\sum_{i,j} X_{ij}\log(X_{ij})$ denotes the generalization of the Shannon entropy over bistochastic matrices.

Proof. **(i)** We follow a procedure similar to the one in Ref. [15]. We make use of the Hilbert-Schmidt inner product $\langle A, B\rangle := \mathrm{Tr}\left(A^\dagger B\right)$ over the space $\mathcal{B}(\mathcal{H})$ of bounded linear operators over $\mathcal{H}$. Starting from Eq. (5) with $c_B = c_B^{(2)}$, we get

$$\begin{aligned} C_B^{(2)}(\mathcal{U}) &= \frac{1}{d}\sum_i \|(\mathcal{I}-\mathcal{D}_B)\mathcal{U}P_i\|_2^2 \\ &= \frac{1}{d}\sum_i \langle(\mathcal{I}-\mathcal{D}_B)\mathcal{U}P_i, (\mathcal{I}-\mathcal{D}_B)\mathcal{U}P_i\rangle \\ &= \frac{1}{d}\sum_i \left(\|\mathcal{U}P_i\|_2^2 - \|\mathcal{D}_B\mathcal{U}P_i\|_2^2\right) \ , \end{aligned}$$

where we have used the fact that the dephasing superoperator $\mathcal{D}_B \in \mathcal{B}(\mathcal{B}(\mathcal{H}))$ is self-adjoint $\mathcal{D}_B^\dagger = \mathcal{D}_B$ with respect to the Hilbert-Schmidt inner product, as well as a projection $\mathcal{D}_B^2 = \mathcal{D}_B$. Unitary invariance of the 2-norm implies $\|\mathcal{U}P_i\|_2^2 = 1$. Using the definition Eq. (2), a straightforward calculation gives

$$C_B^{(2)}(\mathcal{U}) = 1 - \frac{1}{d}\sum_{i,j}(X_\mathcal{U})_{ji}^2 \tag{A1}$$

which reduces to the claimed result.

(ii) Let us denote the Shannon entropy of a probability vector as $H(\boldsymbol{p}) := -\sum_i p_i \log(p_i)$. Since $S(\mathcal{U}P_i) = S(P_i) = 0$, Eq. (5) with $c_B = c_B^{(\mathrm{rel})}$ gives

$$\begin{aligned} C_B^{(\mathrm{rel})}(\mathcal{U}) &= \frac{1}{d}\sum_i S(\mathcal{D}_B\mathcal{U}P_i) \\ &= \frac{1}{d}\sum_i S\left(\sum_j (X_\mathcal{U})_{ji} P_j\right) \\ &= \frac{1}{d}\sum_i H\left(\left\{(X_\mathcal{U})_{ji}\right\}_j\right) \\ &= H(X_\mathcal{U}) \ . \end{aligned}$$

■

Proposition 2. Let $H = \sum_i E_i |\phi_i\rangle\langle\phi_i|$ be a non-degenerate Hamiltonian.

(i) For any state $|\psi\rangle$,

$$\mathcal{P}_\psi = c_B^{(2)}(|\psi\rangle\langle\psi|) \ , \tag{12}$$

where $B = \{|\phi_i\rangle\langle\phi_i|\}_i$ is the eigenbasis of the Hamiltonian.

(ii) Denote the escape probability averaged over a set of orthonormal states $B' = \{|i\rangle\langle i|\}_{i=1}^d$ as

$$\mathcal{P}_{B'} := \frac{1}{d}\sum_{i=1}^d \mathcal{P}_i \ . \tag{13}$$

Then,

$$\mathcal{P}_{B'} = C_B^{(2)}(\mathcal{V}) = C_{B'}^{(2)}(\mathcal{V}^\dagger) \ , \tag{14}$$

where $B = \{|\phi_i\rangle\langle\phi_i|\}_i$ is the eigenbasis of the Hamiltonian and $V = \sum_i |i\rangle\langle\phi_i|$ is the intertwiner between B and B'.

Proof. **(i)** The key observation is that the dephasing superoperator $\mathcal{D}_B$ arises as the (infinite) time average of the Schrödinger evolution $\mathcal{U}_t(\cdot) = e^{-itH}(\cdot)e^{itH}$, namely $\overline{\mathcal{U}_t} = \mathcal{D}_B$. Using the Hilbert-Schmidt inner product over $\mathcal{B}(\mathcal{H})$ (see proof of Proposition 1) and setting $\Pi_\psi = |\psi\rangle\langle\psi|$, we get

$$\begin{aligned} \mathcal{P}_\psi &= 1 - \overline{\mathrm{Tr}\left(\Pi_\psi\,\mathcal{U}_t(\Pi_\psi)\right)} = 1 - \mathrm{Tr}\left(\Pi_\psi\,\mathcal{D}_B(\Pi_\psi)\right) \\ &= 1 - \langle\Pi_\psi, \mathcal{D}_B\Pi_\psi\rangle = 1 - \langle\mathcal{D}_B\Pi_\psi, \mathcal{D}_B\Pi_\psi\rangle \\ &= \langle(\mathcal{I}-\mathcal{D}_B)\Pi_\psi, (\mathcal{I}-\mathcal{D}_B)\Pi_\psi\rangle \\ &= \|(\mathcal{I}-\mathcal{D}_B)\Pi_\psi\|_2^2 = c_B^{(2)}(\Pi_\psi) \ . \end{aligned}$$

(ii) The first equality of Eq. (13) follows by combing part (i) of the Proposition with Eq. (5). For the second equality, from the unitary invariance of the 2-norm, we have

$$\begin{aligned} C_B^{(2)}(\mathcal{V}) &= \frac{1}{d}\sum_i \|(\mathcal{I}-\mathcal{D}_B)\,|i\rangle\langle i|\|_2^2 \\ &= \frac{1}{d}\sum_i \left\|\mathcal{V}^\dagger(\mathcal{I}-\mathcal{D}_{B'})\mathcal{V}\,(|i\rangle\langle i|)\right\|_2^2 \\ &= \frac{1}{d}\sum_i \|(\mathcal{I}-\mathcal{D}_{B'})\mathcal{V}\,(|i\rangle\langle i|)\|_2^2 \\ &= C_{B'}^{(2)}(\mathcal{V}) \ . \end{aligned}$$

However, notice that $X_{\mathcal{V}^\dagger} = X_\mathcal{V}^T$ which from Eq. (7) implies $C_{B'}^{(2)}(\mathcal{V}^\dagger) = C_{B'}^{(2)}(\mathcal{V})$. ■

Appendix B: Inverse participation ratio, effective dimension, and Loschmidt echo

For a non-degenerate Hamiltonian $H = \sum_i E_i |\phi_i\rangle\langle\phi_i|$, the escape probability $\mathcal{P}_\psi$ is directly connected with the second Participation Ratio of $|\psi\rangle$ over the Hamiltonian eigenbasis $\mathrm{PR}_2 := \sum_i |\langle\phi_i|\psi\rangle|^4$ as $\mathcal{P}_\psi = 1 - \mathrm{PR}_2$.

The second Participation ratio, in turn, is intimately connected to two other quantities of physical interest in the study of equilibration and thermalization, namely the *effective dimension* and the *Loschmidt echo* [19, 20]. The effective dimension of a quantum state is defined as its inverse purity,

$$d^{\mathrm{eff}}(\rho) = \frac{1}{\mathrm{Tr}[\rho^2]}, \tag{B1}$$

which intuitively corresponds to the number of pure states that contribute to the (in general) mixed state ρ. Given a non-degenerate Hamiltonian, it is easy to

show that the effective dimension of the (infinite) time-averaged state is equal to the inverse of the second Participation ratio, that is,

$$d^{\text{eff}}(\overline{\rho}) = \frac{1}{\text{Tr}\left(\overline{\rho}^2\right)} = \frac{1}{\sum_i |\langle\phi_i|\psi\rangle|^4} = \frac{1}{\text{PR}_2}, \tag{B2}$$

where $\rho = |\psi\rangle\langle\psi|$.

Recall that the Loschmidt echo is defined as the overlap between the initial state $|\psi\rangle$ and the state after time t,

$$\mathcal{L}_t := \left|\langle\psi|e^{-iHt}|\psi\rangle\right|^2, \tag{B3}$$

the infinite time-average of which can be identified with the *return probability* of the state $|\psi\rangle$. Then, in the non-degenerate case, the time-averaged Loschmidt echo is related to the second Participation ratio and the effective dimension as

$$\overline{\mathcal{L}_t} = \text{PR}_2 = \frac{1}{d^{\text{eff}}(\overline{\rho})}. \tag{B4}$$

For a more detailed exposition, see [42].

Appendix C: CGP in the Anderson model for the degenerate case $W = 0$

The spectrum of Anderson Hamiltonian Eq. (15) for the disorder-free case is degenerate, hence the intertwiner $\mathcal{V}_{W=0}$ between the position and Hamiltonian eigenbases is not uniquely defined. Nevertheless, we show here that the behavior of the quantities $C_B^{(2)}(\mathcal{V}_{W=0})$ and $C_B^{(\text{rel})}(\mathcal{V}_{W=0})$ in the thermodynamic limit is independent of the specific choice of the Hamiltonian eigenbasis, namely $C_B^{(2)}(\mathcal{V}_{W=0}) \to 1$ while $C_B^{(\text{rel})}(\mathcal{V}_{W=0}) \sim \log(L)$ for $L \to \infty$.

The spectrum of the Hamiltonian is $\left\{2\cos\left(\frac{2\pi j}{L}\right)\right\}_{j=0}^{L-1}$, hence there are n_L distinct two-dimensional degenerate subspaces, where $n_L = (L-2)/2$ for L even and $n_L = (L-1)/2$ for L odd. Invoking the Fourier eigenbasis

$$|\phi_k\rangle = \frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} \exp\left(-i\frac{2\pi jk}{L}\right)|j\rangle \tag{C1}$$

as reference, the general eigenbasis of $H_{W=0}$ may differ from basis (C1) as

$$|\phi'_k\rangle = e^{i\gamma_k}\left(e^{i\alpha_k}\cos(\theta_k)\,|\phi_k\rangle + e^{i\beta_k}\sin(\theta_k)\,|\phi_{L-k}\rangle\right) \tag{C2a}$$

$$|\phi'_{L-k}\rangle = e^{i\gamma_k}\left(-e^{i\beta_k}\sin(\theta_k)\,|\phi_k\rangle + e^{-i\alpha_k}\cos(\theta_k)\,|\phi_{L-k}\rangle\right) \tag{C2b}$$

for $k = 1, \ldots, n_L$, where the angles $\{\alpha_k, \beta_k, \gamma_k, \theta_k\}$ specify the (unitary) transformation within the kth two-fold degenerate subspace.

A straightforward calculation gives

$$|\langle l|\phi'_k\rangle|^2 = \left|\langle l|\phi'_{L-k}\rangle\right|^2 = \\ = \frac{1}{L}\left[1 + \cos\left(\frac{2(L-2k)l\pi}{L} + \alpha_k - \beta_k\right)\sin(2\theta_k)\right]. \tag{C3}$$

from which one can directly see that the possible Hamiltonian eigenbases differ in the sum $\sum_{i,j}(X_{\mathcal{U}})_{ji}^2$ at most of an order 1 term. Hence, from Eq. (A1) it follows that any such contribution vanishes at the thermodynamic limit, yielding $C_B^{(2)}(\mathcal{V}_{W=0}) \to 1$.

For $C_B^{(\text{rel})}(\mathcal{V}_{W=0})$, we first invoke the standard inequality between the Shannon entropy and the purity $H(\{p_i\}) \geq -\log\left(\sum_i p_i^2\right)$ (following from the monotonicity of the Rényi entropies [43]). By the use of Eq. (C3), the purity of the probability distribution $\left\{|\langle l|\phi'_k\rangle|^2\right\}_{l=1}^{L}$ is

$$\sum_{l=1}^{L} |\langle l|\phi'_k\rangle|^4 = \frac{2 + \sin^2(2\theta_k)}{2L},$$

therefore the previous inequality implies

$$H\left(\left\{|\langle l|\phi'_k\rangle|^2\right\}_l\right) \geq \log L - \log\left(\frac{2 + \sin^2(2\theta_k)}{2}\right).$$

Finally, this implies by Eq. (8) that $C_B^{(\text{rel})}(\mathcal{V}_{W=0})$ diverges logarithmically with L for any choice of the Hamiltonian eigenbasis.

Appendix D: Derivation of Eqs. (21)

In this section we show how using the ansatz $(X_{\mathcal{V}_W})_{ij} = c_j \exp\left(-|i - \alpha_j|/\xi_j\right)$, one can derive Eqs. (21).

Assuming periodic boundary conditions as in the main text, and since $\sum_i X_{ij} = 1$, the coefficients c_j can be expressed for $L \gg 1$ as

$$(c_j)^{-1} \approx 2\sum_{x=0}^{\infty} e^{-x/\xi_j} - 1$$

therefore

$$c_j = \tanh[(2\xi_j)^{-1}]. \tag{D1}$$

From Eq. (7),

$$C_B^{(2)} = 1 - \frac{1}{L}\sum_{i,j} X_{ij}^2 = 1 - \frac{1}{L}\sum_j \frac{\tanh^2[(2\xi_j)^{-1}]}{\tanh(\xi_j^{-1})},$$

which is (21a).

Similarly, from Eq. (8) we have

$$H\left(X_{V_{W>0}}\right) = -\frac{1}{L}\sum_{i,j=1}^{L} c_j e^{-|i-\alpha_j|/\xi_j} \ln\left[c_j e^{-|i-\alpha_j|/\xi_j}\right]$$
$$= -\frac{1}{L}\sum_j \left(\ln c_j - c_j \sum_i e^{-|i-\alpha_j|/\xi_j}\frac{|i-\alpha_j|}{\xi_j}\right) .$$

The sum $\sum_i$ for $L \gg 1$ is

$$\sum_{i=1}^{L} e^{-|i-\alpha_j|/\xi_j}\frac{|i-\alpha_j|}{\xi_j} \approx 2\sum_{x=1}^{\infty} e^{-x/\xi_j}\frac{x}{\xi_j}$$
$$= -\frac{2}{\xi_j}\frac{d}{d(\xi_j)^{-1}}\sum_{x=1}^{\infty} e^{-x/\xi_j} = 2\frac{e^{1/\xi_j}}{\left(e^{1/\xi_j}-1\right)^2\xi_j} .$$

Using Eq. (D1) together with the above, we get to the desired form (21b).

Appendix E: Evaluation of Eq. (21a) for on-site energies following Cauchy distribution

We consider the Hamiltonian (15) with i.i.d. on-site energies ϵ_i, distributed according to the Cauchy distribution

$$f_\Gamma(\epsilon) = \frac{1}{\pi\Gamma}\left[\frac{\Gamma^2}{\epsilon^2+\Gamma^2}\right] . \tag{E1}$$

The localization length $\xi(E,\Gamma)$ can be calculated by invoking the formula due to Thouless [27], which in our notation is

$$\cosh\left(\frac{1}{2\xi(E,\Gamma)}\right) = \frac{\sqrt{(2+E)^2+\Gamma^2}+\sqrt{(2-E)^2+\Gamma^2}}{4} . \tag{E2}$$

To evaluate Eq. (21a) for this model in the thermodynamic limit, we transition to the continuum limit $\frac{1}{L}\sum_j g(E_j) \mapsto \int dE \rho_\Gamma(E) g(E)$. The density of states $\rho_\Gamma(E)$ can be obtained easily from the corresponding resolvent, calculated for the Lloyd model in [26], and Eq. (E2). The resulting integral is numerically evaluated and yields the data plotted in Figure 3.

Appendix F: Coherence-Generating Power and distance in the Grassmannian

Here we present in more detail the underlying differential-geometric structure that is introduced in section V.

Let $\mathcal{H}$ denote the finite dimensional Hilbert space of the quantum system and $\mathcal{B}(\mathcal{H})$ the associate operator algebra. The set $\mathcal{B}(\mathcal{H})$ equipped with the Hilbert-Schmidt scalar product $\langle X, Y\rangle := \mathrm{Tr}\left(X^\dagger Y\right)$ turns into a Hilbert space (the space of Hilbert-Schmidt operators) that we will denote by $\mathcal{H}_{\mathrm{HS}}$. Superoperators $\mathcal{O}$ mapping $\mathcal{H}_{\mathrm{HS}}$ into itself can be then endowed with the following norm

$$\|\mathcal{O}\|_{\mathrm{HS}} := \sqrt{\mathrm{Tr}_{\mathrm{HS}}(\mathcal{O}^\dagger\mathcal{O})} , \tag{F1}$$

where (a) $\mathcal{O}^\dagger$ denotes the Hilbert-Schmidt conjugate of $\mathcal{O}$, i.e., $\langle \mathcal{O}(X), Y\rangle = \langle X, \mathcal{O}^\dagger(Y)\rangle \ \forall X, Y \in \mathcal{H}_{\mathrm{HS}}$. (b) If $\{|i\rangle\}_{i=1}^d$ is any orthonormal basis of $\mathcal{H}$, one defines $\mathrm{Tr}_{\mathrm{HS}}\mathcal{O} := \sum_{i,j=1}^d \langle |i\rangle\langle j|, \mathcal{O}(|i\rangle\langle j|)\rangle$.

As we discussed in the main text, instead of invoking orthonormal sequences of kets $\{|i\rangle\}_{i=1}^d$, it is more convenient to work with sets of orthogonal, rank-1 projection operators $B = \{P_i := |i\rangle\langle i|\}_{i=1}^d$. Let us introduce the space of all such sets over the Hilbert space, which we denote as $\mathcal{M}(\mathcal{H})$. This is essentially the set of all possible orthonormal bases over the Hilbert space once the phase degrees of freedom and ordering have been modded out [10]. The elements $B \in \mathcal{M}(\mathcal{H})$ are in one-to-one correspondence with the set of dephasing super-operators, i.e., the map $B \mapsto \mathcal{D}_B$ (defined in Eq. (2)) is injective. Given a $B \in \mathcal{M}_d$, the corresponding set of B-diagonal operators is

$$\mathcal{A}_B := \mathrm{Span}\,\{P_i\}_{i=1}^d \subset \mathcal{H}_{\mathrm{HS}} , \tag{F2}$$

which is also the range of the B-dephasing superoperator $\mathcal{D}_B$. One can see that Eq. (F2) actually defines a Maximally Abelian Sub-Algebra (MASA) of $\mathcal{H}_{\mathrm{HS}}$; moreover it can be proven that the set of MASAs of $\mathcal{H}_{\mathrm{HS}}$ can be identified with $\mathcal{M}(\mathcal{H})$ (see [10] for a proof). In this way, the set $\mathcal{M}(\mathcal{H})$ can be now seen as a subset of the Grassmannian manifold of d-dimensional subspaces of $\mathcal{H}_{\mathrm{HS}}$. The advantage of this approach is that $\mathcal{M}(\mathcal{H})$ directly inherits the natural metric structure of the Grassmannian

$$D(\mathcal{A}_B, \mathcal{A}_{B'}) := \|\mathcal{D}_B - \mathcal{D}_{B'}\|_{\mathrm{HS}} . \tag{F3}$$

We will now connect these concepts to the 2-CGP of unitary quantum maps.

From its definition, $C_B^{(2)}(\mathcal{U})$ seems to capture some notion of separation between the sets $B = \{P_i\}_{i=1}^d$ and $B' = \{\mathcal{U}(P_i)\}_{i=1}^d$. In fact, the B-coherence generating power of a unitary map $\mathcal{U}$ is proportional to the (square of the) Grassmannian distance between the input B-diagonal algebra $\mathcal{A}_B$ and its image under $\mathcal{U}$ [10]. Formally:

$$C_B^{(2)}(\mathcal{U}) = \frac{1}{2d} D(\mathcal{A}_B, \mathcal{U}(\mathcal{A}_B))^2. \tag{F4}$$

where the distance function D is given by (F3). The maximum of this function i.e., $\max_{\mathcal{U}} C_B^{(2)}(\mathcal{U}) = 1 - 1/d$ is achieved for unitary operators $\mathcal{U}$ that connected mutually unbiased bases, namely $|\langle i|U|j\rangle| = 1/d$ $(\forall i, j)$, and corresponds to a maximum distance over $\mathcal{M}(\mathcal{H})$ given by $D^{\max} = \sqrt{2(d-1)}$.

It is important to stress that, in the light of Proposition 2, the Grassmannian distance between MASAs is

endowed with a physical meaning in the context of quantum mechanics.

We now turn to establish a connection between the differential structure of $\mathcal{M}(\mathcal{H})$, as induced by the distance function (F3), and MBL. One has the natural Riemannian metric over the Grassmannian

$$ds^2 = D(\Pi, \Pi + d\Pi)^2 = \mathrm{Tr}(d\Pi^2) \tag{F5}$$

(Π denote the projectors over the d-dimensional subspaces comprising the Grassmannian). The latter, in view of Eq. (F4), has in turn the physical interpretation as the $C_B^{(2)}$ of the unitary associated with an infinitesimal transformation $\{|\phi_i(\lambda)\rangle\}_{i=1}^d \mapsto \{|\phi_i(\lambda + d\lambda)\rangle\}_{i=1}^d$. The form of the metric (28) follows directly by the calculation of Proposition 6 in [10].